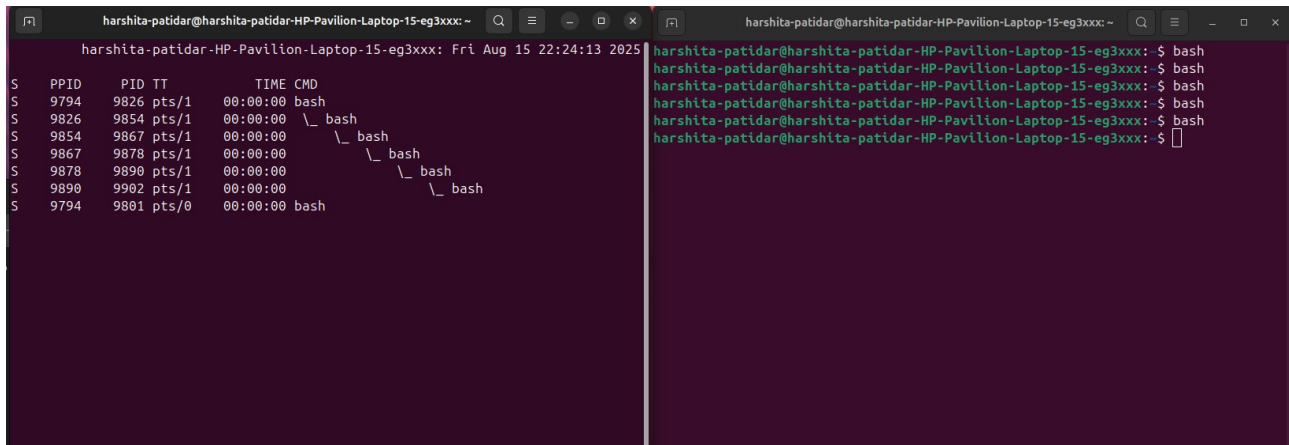


Before Kill:

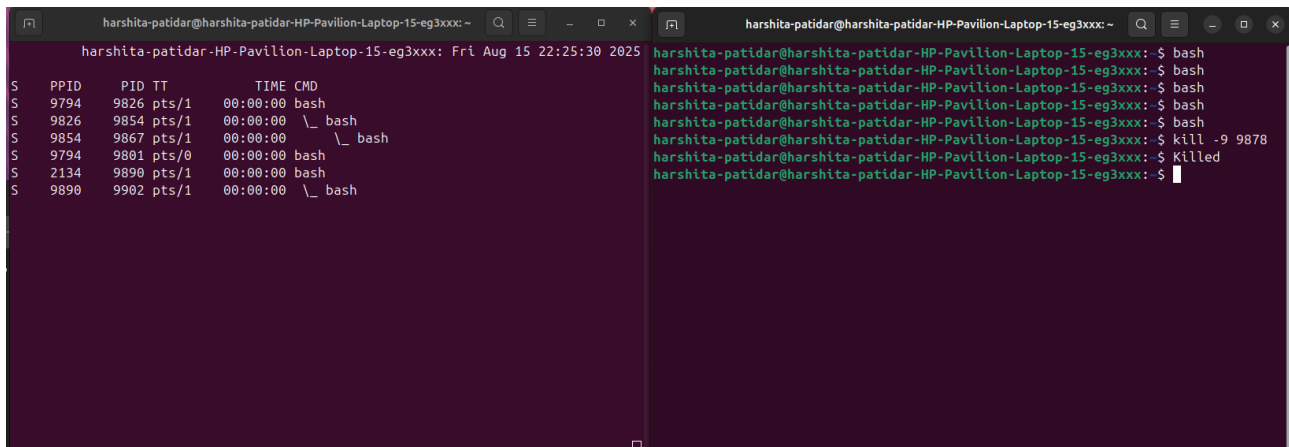


The image shows two terminal windows. The left window displays the output of the 'ps' command, showing a tree of processes. The right window shows the command prompt with several 'bash' commands entered, demonstrating the process tree structure.

```
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx: ~  
harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx: Fri Aug 15 22:24:13 2025  
S  PPID  PID TT      TIME CMD  
S  9794  9826 pts/1    00:00:00 bash  
S  9826  9854 pts/1    00:00:00 \_ bash  
S  9854  9867 pts/1    00:00:00 \_ bash  
S  9867  9878 pts/1    00:00:00 \_ bash  
S  9878  9890 pts/1    00:00:00 \_ bash  
S  9890  9902 pts/1    00:00:00 \_ bash  
S  9794  9881 pts/0    00:00:00 bash
```

```
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx: ~  
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx: $ bash  
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx: $ bash  
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx: $ bash  
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx: $ bash  
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx: $ bash  
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx: $
```

After kill:



The image shows two terminal windows. The left window displays the output of the 'ps' command, showing the process tree after killing process 9878. The right window shows the command prompt with several 'bash' commands entered, demonstrating the process tree structure after the kill command.

```
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx: ~  
harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx: Fri Aug 15 22:25:30 2025  
S  PPID  PID TT      TIME CMD  
S  9794  9826 pts/1    00:00:00 bash  
S  9826  9854 pts/1    00:00:00 \_ bash  
S  9854  9867 pts/1    00:00:00 \_ bash  
S  9794  9881 pts/0    00:00:00 bash  
S  2134  9890 pts/1    00:00:00 bash  
S  9890  9902 pts/1    00:00:00 \_ bash
```

```
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx: ~  
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx: $ bash  
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx: $ bash  
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx: $ bash  
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx: $ bash  
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx: $ bash  
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx: $ kill -9 9878  
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx: $ Killed  
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx: $
```

Observation:

The **init** process adopted the orphaned child process.

It is the first process started by the kernel at boot and is responsible for reaping zombie processes and adopting orphaned processes to ensure proper process cleanup.